

New issue



[Bug]: Prefix caching ignores visual input, causing incorrect multimodal outputs under concurrency #20261

Open

#36622

Labels

bug

unstable



Richar-Du opened on Jun 30, 2025



Your current environment

► The output of `python collect_env.py`



Describe the bug

When serving the qwen2.5-vl multimodal model using vllm serve, I noticed that enabling `--enable-prefix-caching` leads to incorrect outputs (e.g., repeated, truncated, or garbled tokens) under high concurrency or heavy asynchronous usage. The Prefix cache hit rate is around 40%. All the requests use identical text prompts but slightly different images (The images are mazes with different structures).

Disabling prefix caching (`--no-enable-prefix-caching`) resolves the issue. I wonder if the cache key only considers text prompts and does not take image content or image-derived embeddings into account? Or although the images are different mazes, the embedding may be quite similar, leading to the incorrect cache matching?

Before submitting a new issue...

- ☒ Make sure you already searched for relevant issues, and asked the chatbot living at the bottom right corner of the [documentation page](#), which can answer lots of frequently asked questions.

4



Richar-Du added **bug** on Jun 30, 2025



DarkLight1337 on Jul 1, 2025 · edited by DarkLight1337

Edits ▼

Member



Could you provide an example script to reproduce the issue?



Godricly on Jul 7, 2025



```
--disable-mm-preprocessor-cache --no-enable-prefix-caching --disable-chunked-mm-input
```

try if this help mitigating your problem. if the vllm crashes due to memory issue, try adding `--enforce-eager`. this helped me in my scenario.



2



Richar-Du on Jul 11, 2025

Author



@DarkLight1337 I noticed the same issue others have encountered here: [QwenLM/Qwen3-VL#1093](#). You might want to try the example script provided in that thread. In my case, the model also generated the odd word "addCriterion".

@Godricly Thank you for your suggestion, but it didn't work for me.



ILikeIneine on Jul 24, 2025 · edited by ILikeIneine

Edits ▼

Contributor



leads to incorrect outputs (e.g., repeated, truncated, or garbled tokens) under high concurrency or heavy asynchronous usage. The Prefix cache hit rate is around 40%.

Get all the problems while testing qwen2.5-vl-7b with the same phenomenon.

```
--no-enable-prefix-caching
```

After disable prefix caching it goes well with correct result.



2



AILaborer on Aug 7, 2025



I encountered a similar issue.

When using the QWEN3 text generation model on an RTX 3090, high concurrency leads to chaotic outputs that are completely different from those produced without parallelism.(Strangely, I don't encounter any issues on the RTX 5060 Ti. Both setups are deployed using Docker, with an identical environment—only the GPU visible to the container differs.)

Adding `--no-enable-prefix-caching` resolves the issue.



Godricly on Aug 8, 2025 · edited by Godricly

Edits ▾ ⋮

The latest v0.10 narrows the gap caused by `--no-enable-prefix-caching` when testing my model(qwenvl-7b). And `--enforce-eager` with `--no-enable-prefix-caching` gives the worst result.



DarkLight1337 on Aug 8, 2025

Member ⋮

cc [@ywang96](#) for visibility



Godricly mentioned this on Aug 15, 2025



`--disable-mm-preprocessor-cache` now expects argument #22900



Godricly on Aug 15, 2025

⋮

Can you guys try hard coding `enable_chunked_prefill` to `False` to see if this mitigate your problem? This was turn on by default in V1 and unable to disable through `--no-enable-chunked-prefill` argument.

[vllm/vllm/engine/arg_utils.py](#)

Lines 1575 to 1578 in 49252cf

```
1575     if model_config.runner_type != "pooling":
1576         self.enable_chunked_prefill = True
1577         if self.enable_prefix_caching is None:
1578             self.enable_prefix_caching = True
```



SCBoy1007 added a commit that references this issue on Oct 24, 2025

Fix vLLM multimodal cache bug and add comprehensive debug logging ...

277e05e



github-actions bot on Nov 14, 2025 – with [GitHub Actions](#)

⋮

This issue has been automatically marked as stale because it has not had any activity within 90 days. It will be automatically closed if no further activity occurs within 30 days. Leave a comment if you feel this issue should remain open. Thank you!

 **github-actions** added **stale** on Nov 14, 2025

 ZeroYujie on Nov 25, 2025

I am currently using vLLM to deploy LLM. Recently, I adjusted to use the same node for inference in the same session, and I also encountered the problem of repeated responses.

 DarkLight1337 on Nov 25, 2025

Member ...

Have you tried `--no-enable-prefix-caching` ?

 **github-actions** added **unstale** and removed **stale** on Nov 26, 2025

 ZeroYujie on Nov 26, 2025

Have you tried `--no-enable-prefix-caching` ?

I believe the prefix cache helps improve inference speed, so it hasn't been disabled yet. I will try disabling it

 DarkLight1337 on Nov 26, 2025

Member ...

Yes it does, but we are trying to isolate the issue here.

 **github-actions** **bot** on Feb 25 – with [GitHub Actions](#)

This issue has been automatically marked as stale because it has not had any activity within 90 days. It will be automatically closed if no further activity occurs within 30 days. Leave a comment if you feel this issue should remain open. Thank you!

 **github-actions** added **stale** and removed **unstale** on Feb 25



ZeroYujie on Mar 6

...

Has the issue with the prefix cache been resolved in version v0.16.0? [@DarkLight1337](#)



DarkLight1337 on Mar 6

Member

...

We haven't managed to repro the issue, so not sure. Do you still get this problem?



github-actions added **unstale** and removed **stale** on Mar 7



AjAnubolu linked a pull request that will close this issue on Mar 10

[🔗 \[Bugfix\] Fix off-by-one in multimodal prefix cache hash boundary check #36622](#)

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Metadata

Assignees

No one assigned

Labels

bug

unstale

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

Relationships

None yet

Development

No branches or pull requests
